

Ex.No. 2: Implementing TDMA Slot Allocation in GSM. Date

```
clc; clear; close all;
```

```
FrameTime = 4.615; Slots = 8;
```

```
SlotDuration = FrameTime/Slots;
```

```
t = 0:SlotDuration:FrameTime;
```

```
figure
```

```
subplot(2,1,1)
```

```
stem(t,ones(size(t)),'filled')
```

```
title('TDMA Time Slot Structure')
```

```
xlabel('Time (ms)')
```

```
ylabel('Slot Level')
```

```
grid on
```

```
subplot(2,1,2)
```

```
plot(t,ones(size(t)),'-o','LineWidth',1.5)
```

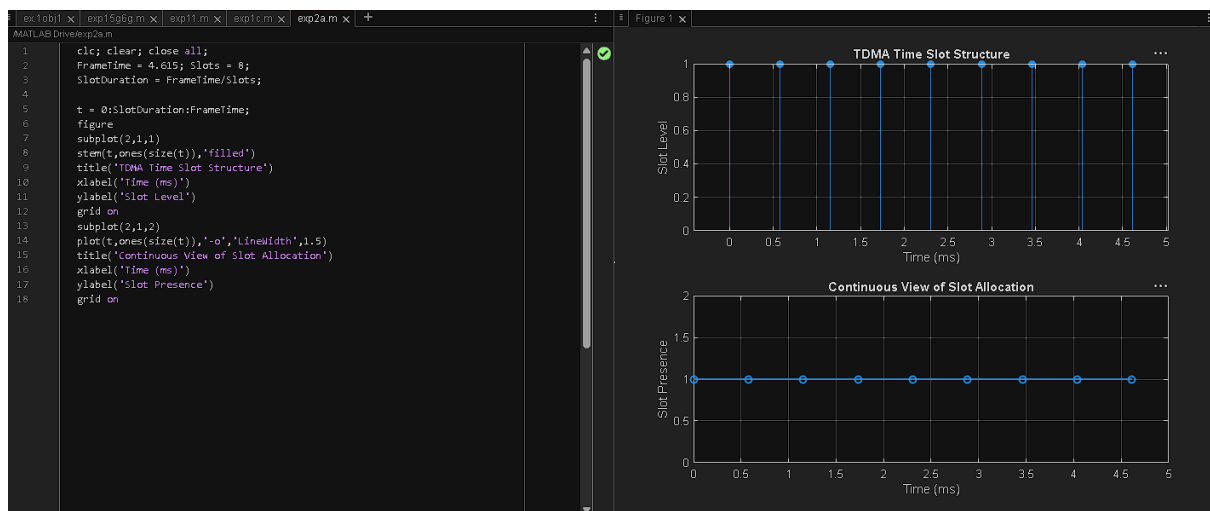
```
title('Continuous View of Slot Allocation')
```

```
xlabel('Time (ms)')
```

```
ylabel('Slot Presence')\
```

```
grid on
```

output:



Objective 2

```
clc; clear; close all;
```

```
TotalSlots = 8;
```

```
AllocatedSlots = 6;
```

```
FreeSlots = TotalSlots - AllocatedSlots;
```

```
Utilization = (AllocatedSlots/TotalSlots)*100;
```

```
disp('Frame Utilization (%)')
```

```
disp(Utilization)
```

```
figure
```

```
% Graph 1: Simple Line Plot (SAFE - no bar function)
```

```
subplot(2,1,1)
```

```
plot([1 2],[AllocatedSlots FreeSlots],'-o','LineWidth',2)
```

```
title('GSM Frame Utilization (Allocated vs Free)')
```

```
xlabel('Slot Type Index (1=Allocated, 2=Free)')
```

```
ylabel('Number of Slots')
```

```
grid on
```

```
% Graph 2: Pie Chart (SAFE)
```

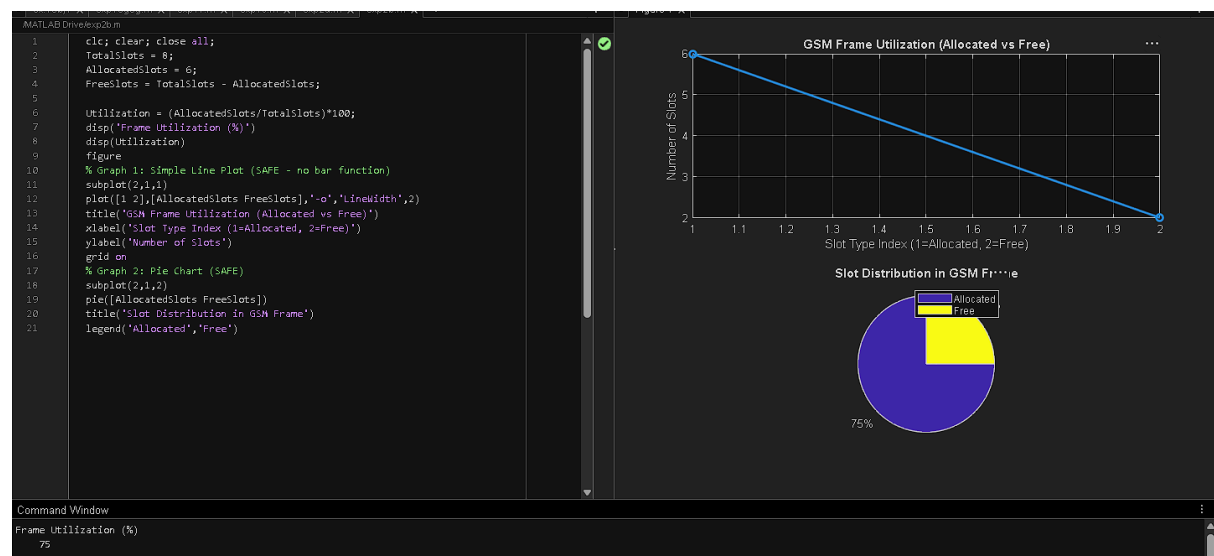
```
subplot(2,1,2)
```

```
pie([AllocatedSlots FreeSlots])
```

```
title('Slot Distribution in GSM Frame')
```

```
legend('Allocated','Free')
```

output



Objective 3

```
clc;
```

```
clear;
```

```

close all;

% Parameters

Users = 1:8;      % Total users

TotalSlots = 5;   % Available TDMA slots

% Slot allocation

Alloc = zeros(1,8); % Initialize allocation

Alloc(1:TotalSlots) = 1; % Assign slots to first 5 users

Blocked = 1 - Alloc; % Remaining users are blocked

% Display allocation table

disp('User Wise Slot Allocation');

T = table(Users', Alloc', ...
    'VariableNames', {'User', 'Status_1_Allocated_0_Blocked'});

disp(T);

% Plot graphs

figure;

% Graph 1: Allocation per user

subplot(2,1,1);

stem(Users, Alloc, 'filled', 'LineWidth', 2);

title('TDMA Slot Allocation per User');

xlabel('User Index');

ylabel('Slot Status');

ylim([-0.2 1.2]);

grid on;

% Graph 2: Allocation vs Blocked Users

subplot(2,1,2);

plot(Users, Alloc, '-o', 'LineWidth', 2);

hold on;

plot(Users, Blocked, '-s', 'LineWidth', 2);

hold off;

title('TDMA Slot Utilization Analysis');

xlabel('User Index');

ylabel('Status');

legend('Allocated', 'Blocked', 'Location', 'best');

ylim([-0.2 1.2]);

```

grid on;

output:

